



- 4 One end of a light elastic string of natural length a and modulus of elasticity $5mg$ is attached to a fixed point O . Two particles, P and Q , of masses m and $4m$ respectively are attached to the other end of the string and they hang vertically in equilibrium. Particle Q is then detached from the string, hence releasing particle P from rest.

Find, in terms of a , the length of the string when the speed of particle P is first equal to $\sqrt{\frac{7}{5}ag}$. [6]

This image shows a full page of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page, typical of notebook or legal stationery. There are no margins, text, or other markings on the page.



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